

15. Active Directory Lab

What is Active Directory (AD)

- Like the name suggests it's a directory
- Phonebook is another example of a directory
- AD is Microsoft's (very popular) directory service
- AD contains details like:
 - Name
 - Email
 - Username
 - Password
 - What resources you can access
 - What computers you can login to
 - What you can do on those computers

Why Active Directory (AD)

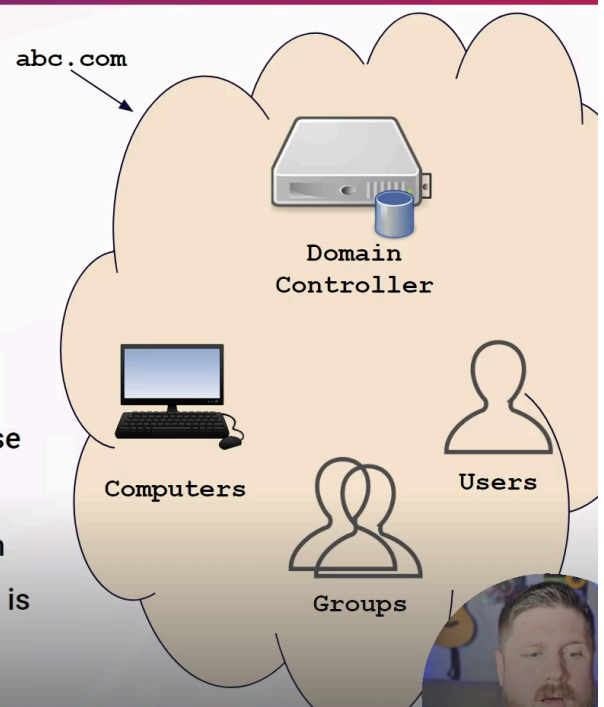
- Centralized IAM for users and resources in an organization
- Ease of administration by having centralized source of truth
- Enhanced security, allows for much more granular controls of who can do and access what
- Single point for backup and recovery

Physical Components of AD

- **Domain Controller**
 - Server (physical, virtual, cloud in Azure)
 - Hosts AD Database
 - Responds to authentication requests
 - Process logins
 - Enforces policies
- **AD Database (NTDS.dit)**
 - Stores the data for the directory
 - Stores log files
 - Structure of the DB is called the Schema

Logical Components of AD

- Logical components of AD are what provide structure to how the environment is laid out
- They allow for organization and management of users and resources
- In AD essentially everything inside of the database is considered an object
- Some objects are also containers in that they can hold other objects, think of them like folders, this is what gives AD its hierarchical structure



Domains

- One of the top level containers is a domain
- Holds a collection of objects (users, computers, groups and more)
- Serves as an administrative boundary for managing the objects inside of it
- Can enforce security and policy settings across all objects inside the domain
- Domains are referred to by their domain name in this example abc.com
- Frequently this will line up with an organizations registered domain

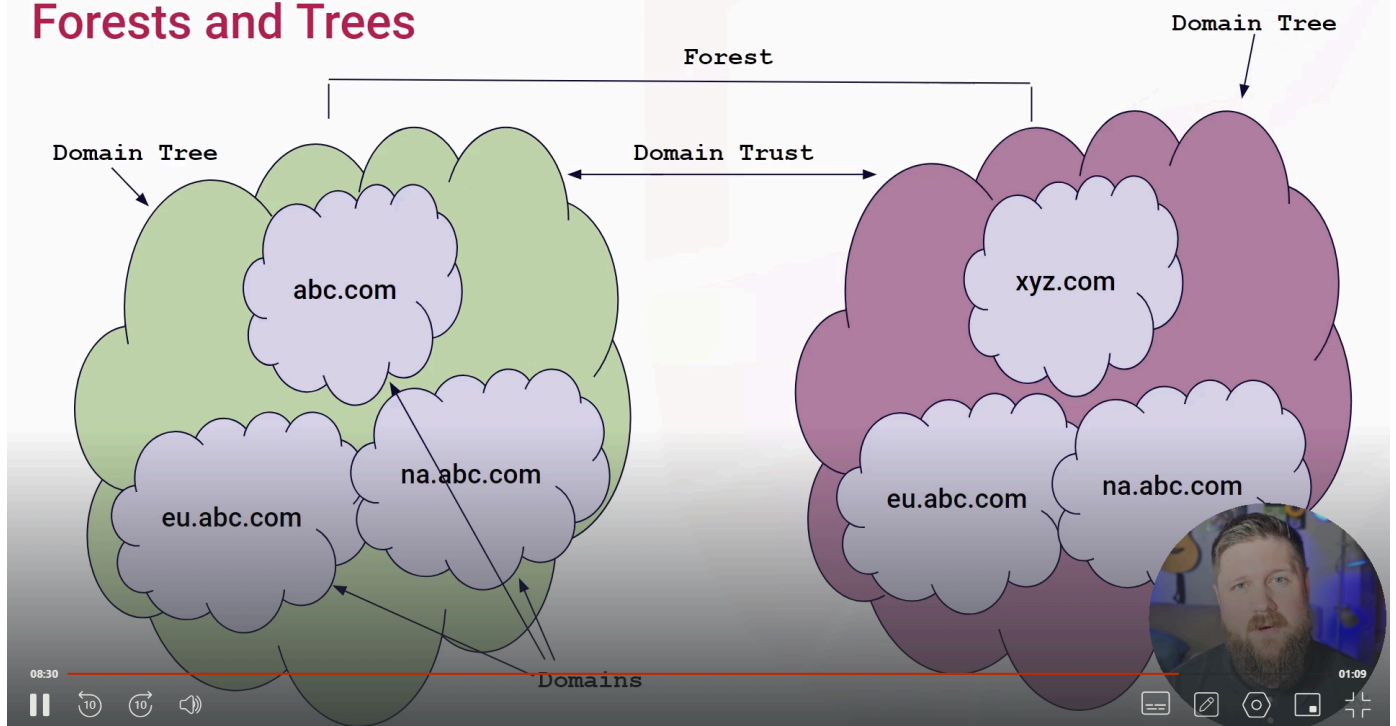
Organizational Units

- A container object inside of domains
- Used to group together other objects
- Lowest level container that can allow administrative privileges over it
- Lowest level container that can have group policies applied to it
- Frequently organizations will split up departments into OUs (Sales, HR, IT, Engineering etc)

Non-Container Objects

- Users
- Computers
- Groups (used for setting access)
- Printers
- Group Policy Objects

Forests and Trees



Install guest additions and then take snapshot after installing Windows Server 2025 in vm.

in the server manager, click Manage and then click add roles and features.

enable active directory services and then install it and then promote to the domain controller and then install it and then restart it.

install AD CS also as a feature to that new Domain Controller.

Adding new users:

Go to tools, click AD users and computers,

go to LAB.local -> Users

Right click LAB.local, and select New -> Organizational Unit.

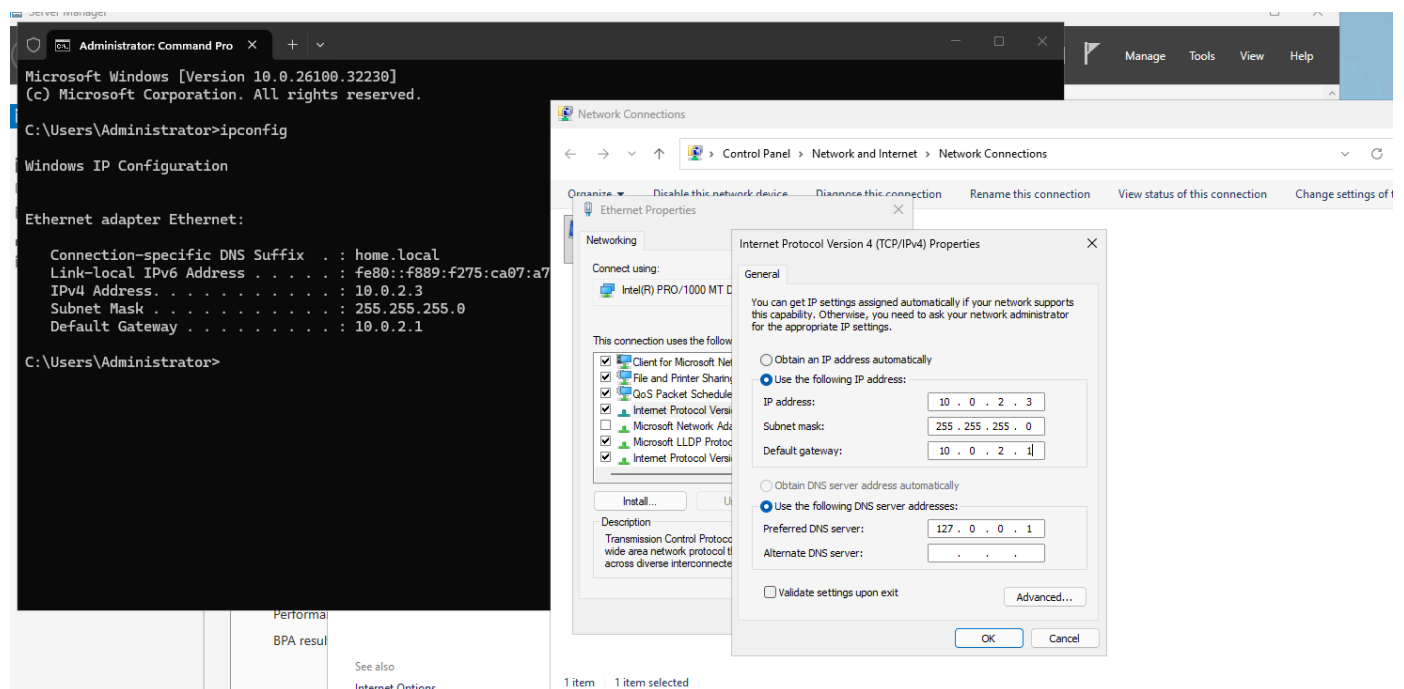
Select the users' groups and drag it to the (Groups) = name of the Organizational Unit.

On the users folder, right click on the empty space and create new user. create 5 users.

in virtual box, creat a network, NAT network, name it ADNetwork. Then for the win 11 ent vm and this server 2025 vm, edit the network settings, put the NAT network as the 'attached to'.

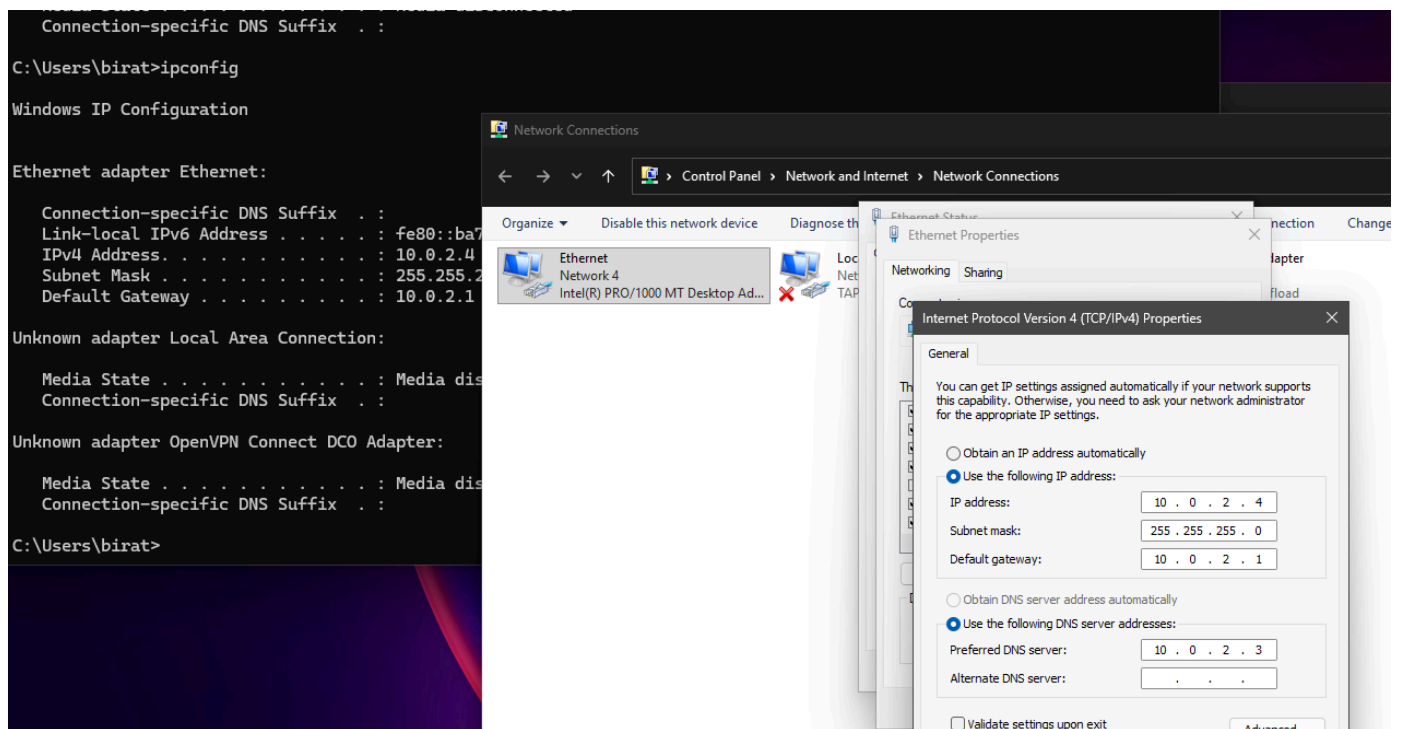
start both windows vms.

go to the adapter settings of server 2025 vm and then select the properties of ethernet. then go to thier properties and edit the ipv4 settings, make use the following ip address and put the same on as the settings of vbox shows for the NAT Network.



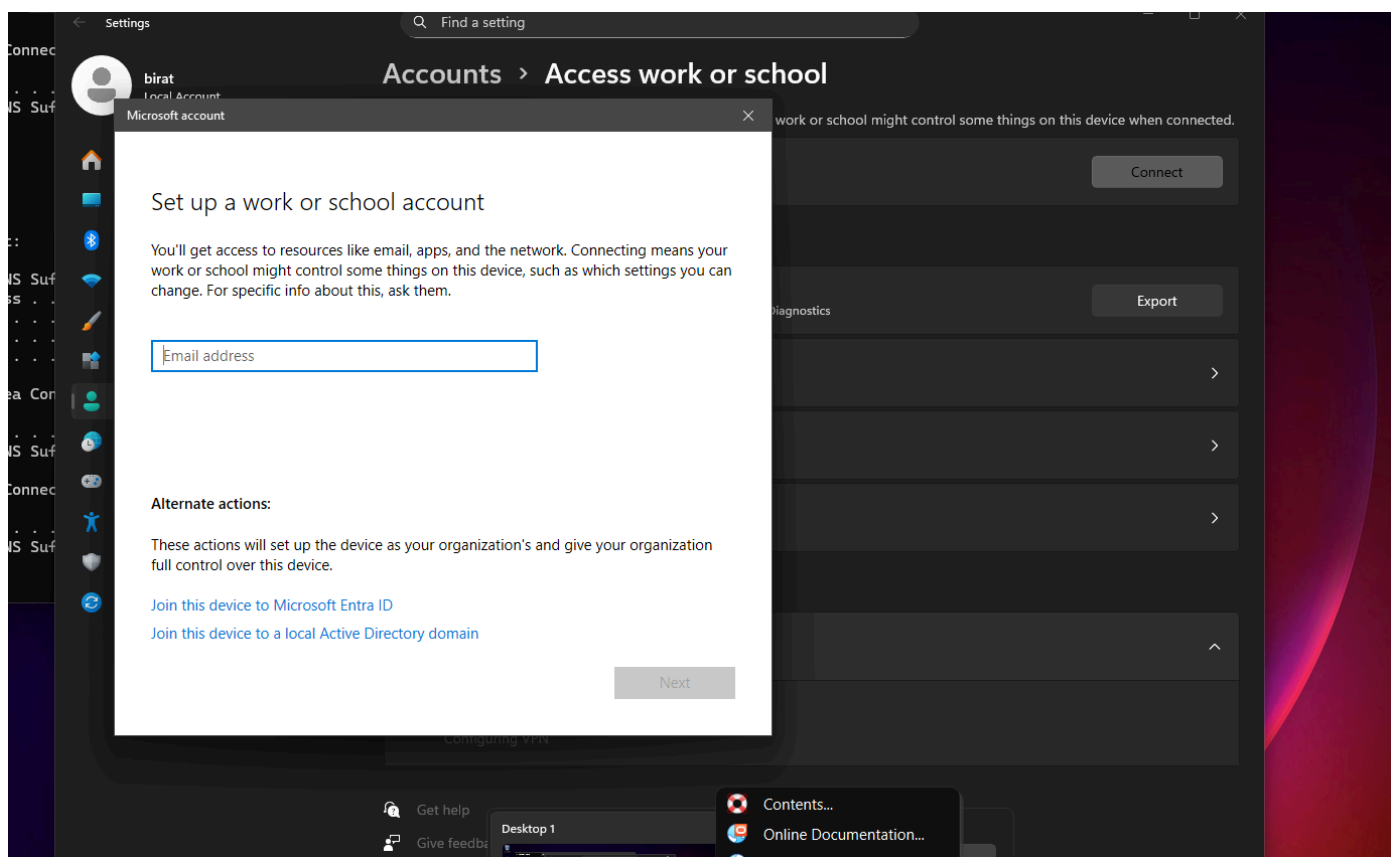
change the name of the win 11 ent vm to WS01.

go change the ipv4 on this one as well, the consecutive ip. and then put the server's ip as the DNS server on this one.



now on the win 11 ent vm, go to 'access work or school' settings.

and click on 'connect' and click on 'join this device to local ad domain'



and then put in the field 'LAB.local'

username = administrator

password = same password as of the local administrator.

now restart and login is rhendricks.

hotmart.com - To exit full screen, pre

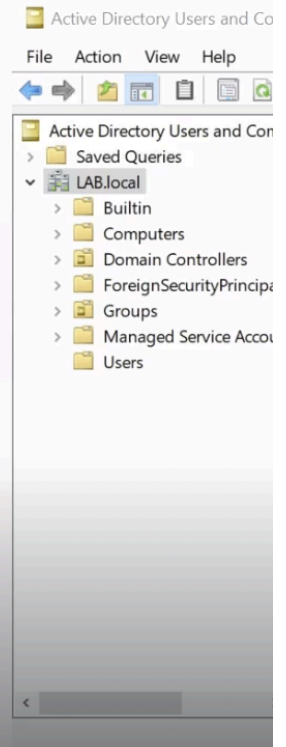
Organizational Unit (OU)

- Container in Active Directory used to organize and administer objects such as users, computers, groups and even other OUs
- OUs are the smallest administrative unit in Active Directory
- Can assign administrative privileges to a user or group for an OU
- Can apply Group Policy Objects to an OU to affect all objects within it
- Not to be confused with “Containers”

now, on the server manager, click the tools, and AD users and computers,
now , go to users, now it should show the name of the win 11 ent vm.

Difference Between OUs and Groups

- Groups are another object in AD
- Groups can be used to assign permissions and access, for example to shared folders
- Groups cannot be used for administration like applying GPOs
- OUs can contains groups, groups cannot contain OUs
- Objects such as users can be part of multiple groups, but only one OU



the layout of the groups and OUs is called a Schema.

now, right click on LAB.local, create new -> Ou

create 3 new OUs, put 5 users in different OUs.

create another OU 'administrators' under IT OU.

and put the admin user on there.

Create a security group inside Engineering OU.

Add the Engineering department users on there and also one member from Management to there.

go to File and Storage Services to create a shared folder.

go to Shares -> Tasks (new Share)

make SMB share quick.

Disable Inheritance, Convert inherited permissions into explicit permissions on this object.

remove the lab users without full control, then add -> Select a principal,
add EngineeringShare.

login using a management user who was also included in that OU administrators on the win 11 ent vm.

now, on the Windows Explorer, put

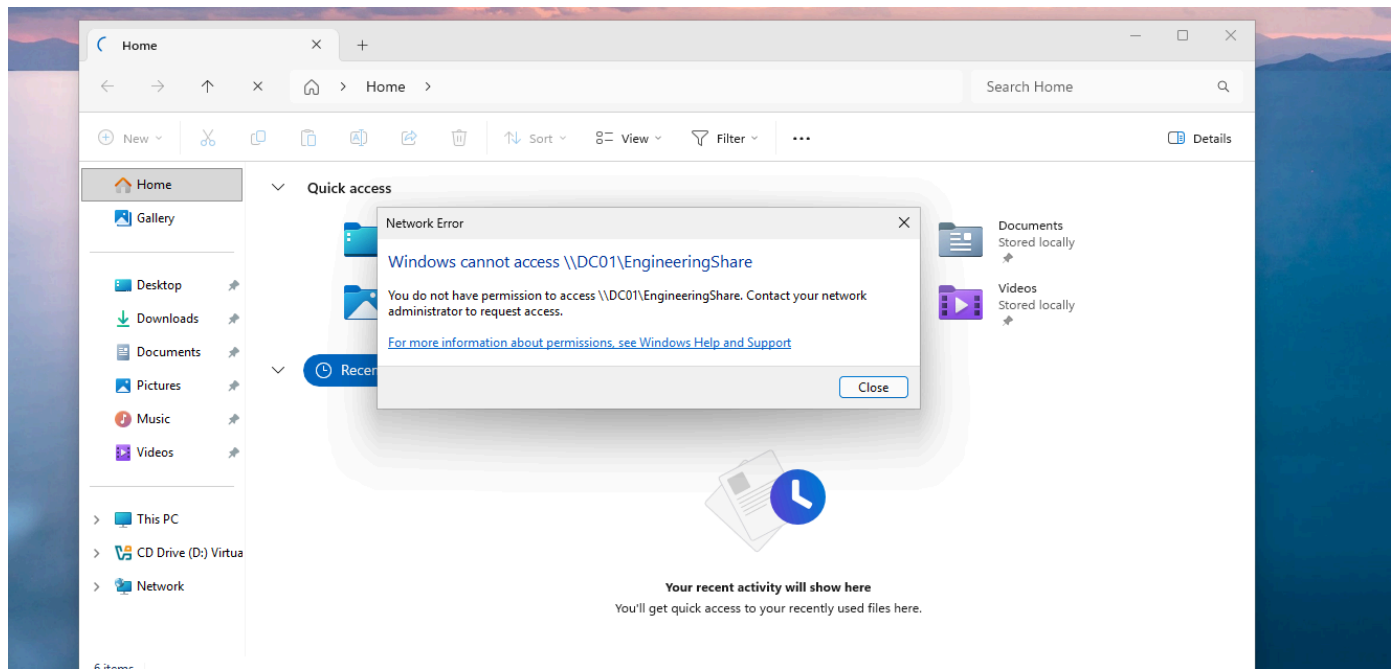
\\DC01\\EngineeringShare, to access the shared folder.

create a text document there.

Map the network drive.

sign out and then sign in again as another management user.

try to access the share folder and it won't allow because this user is not in that group.

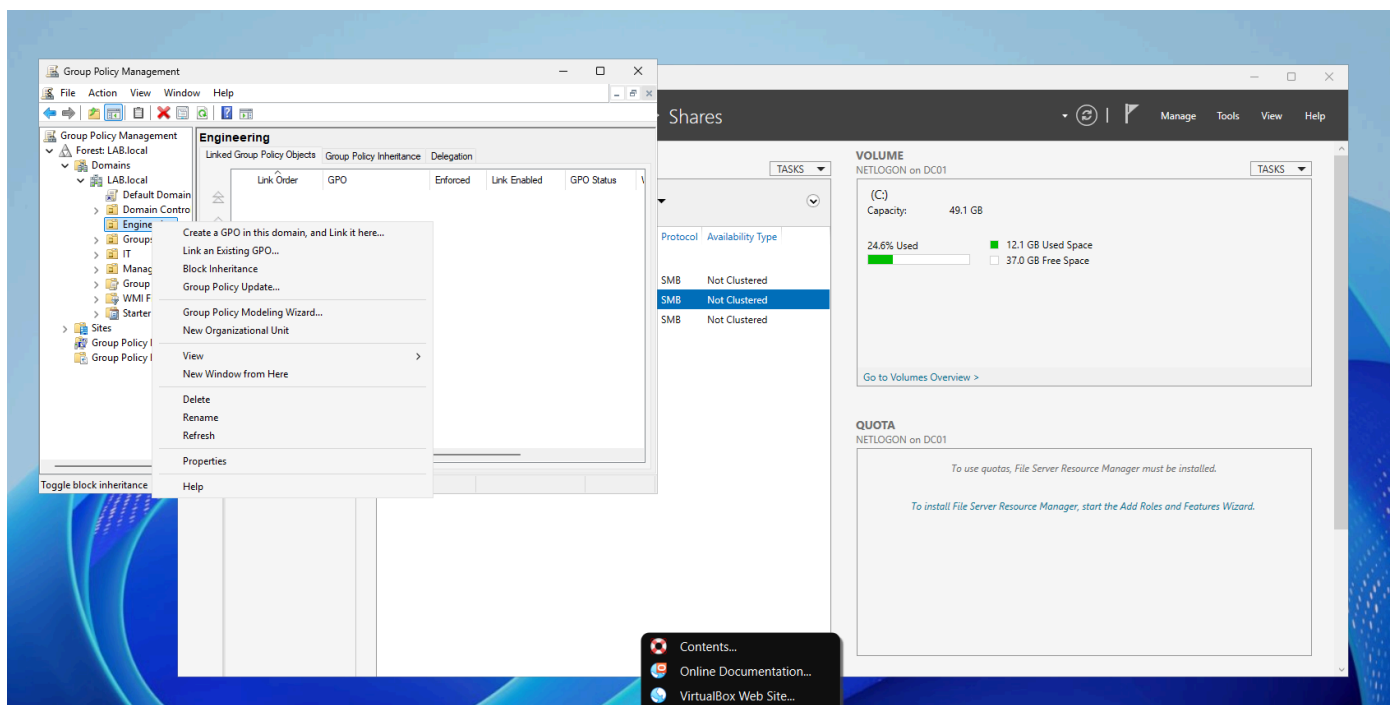


download a wallpaper on the win server 2025.

go to netlogon on shares tab from the file and storage services on the server manager, right click and open share.

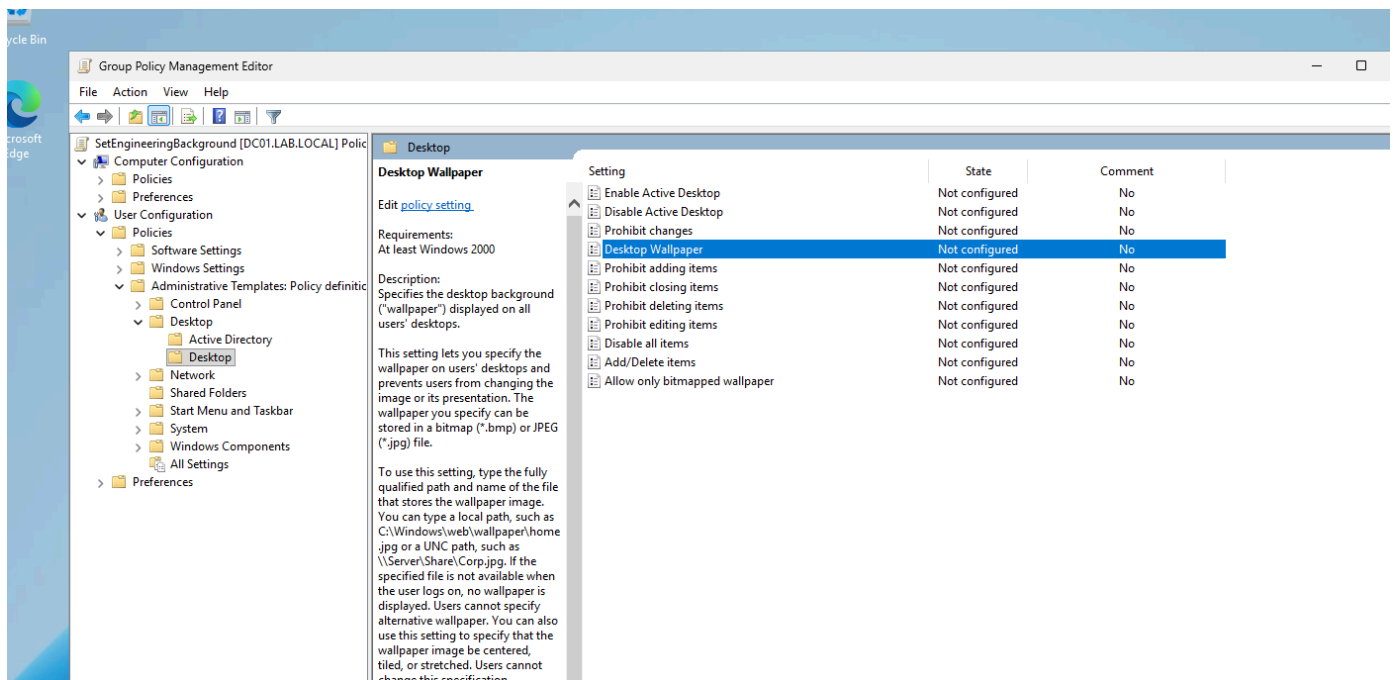
copy that wallpaper to the netlogon folder, and copy that as path.

go to tools in the Server Manager, and go to Group Policy Management.



click the first option on the dropdown. name it and hit ok.

right click and edit that.

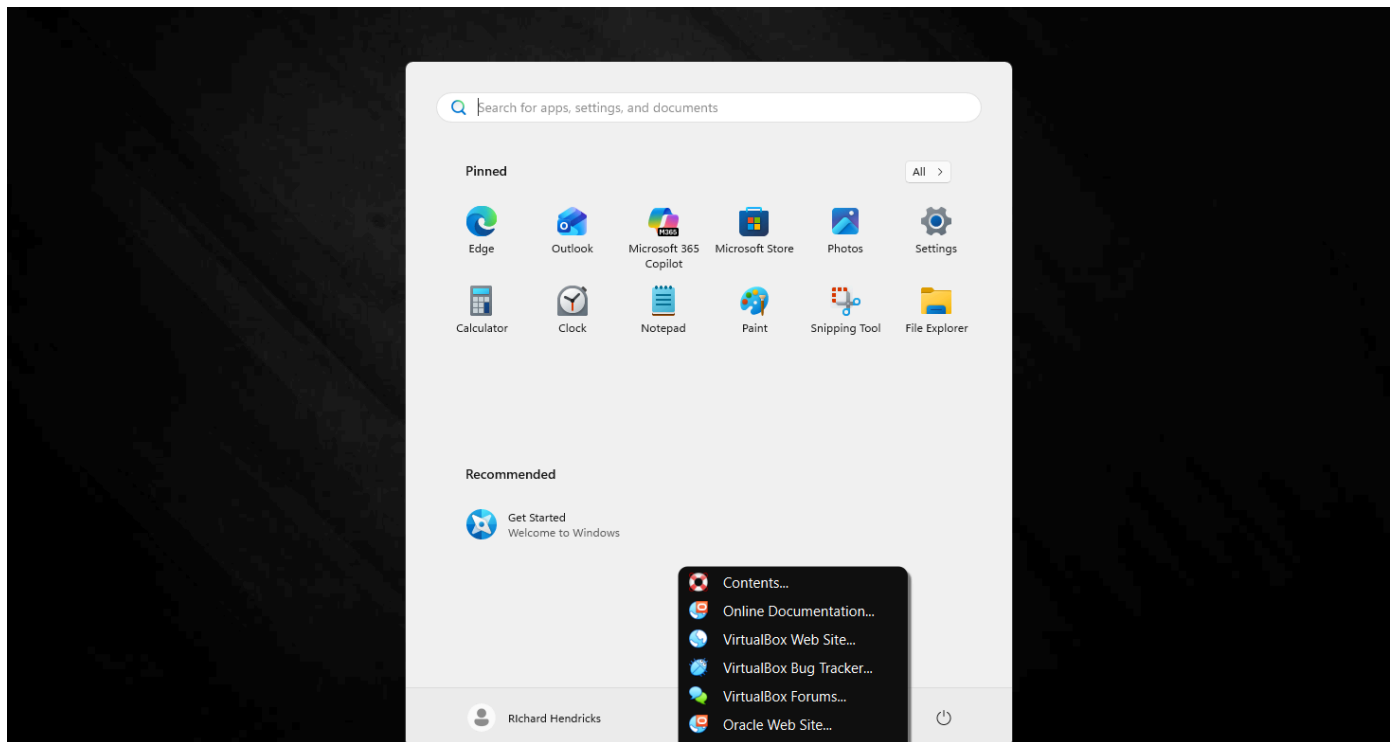


double click desktop wallpaper and select enable.

paste the path on the Wallpaper name box.

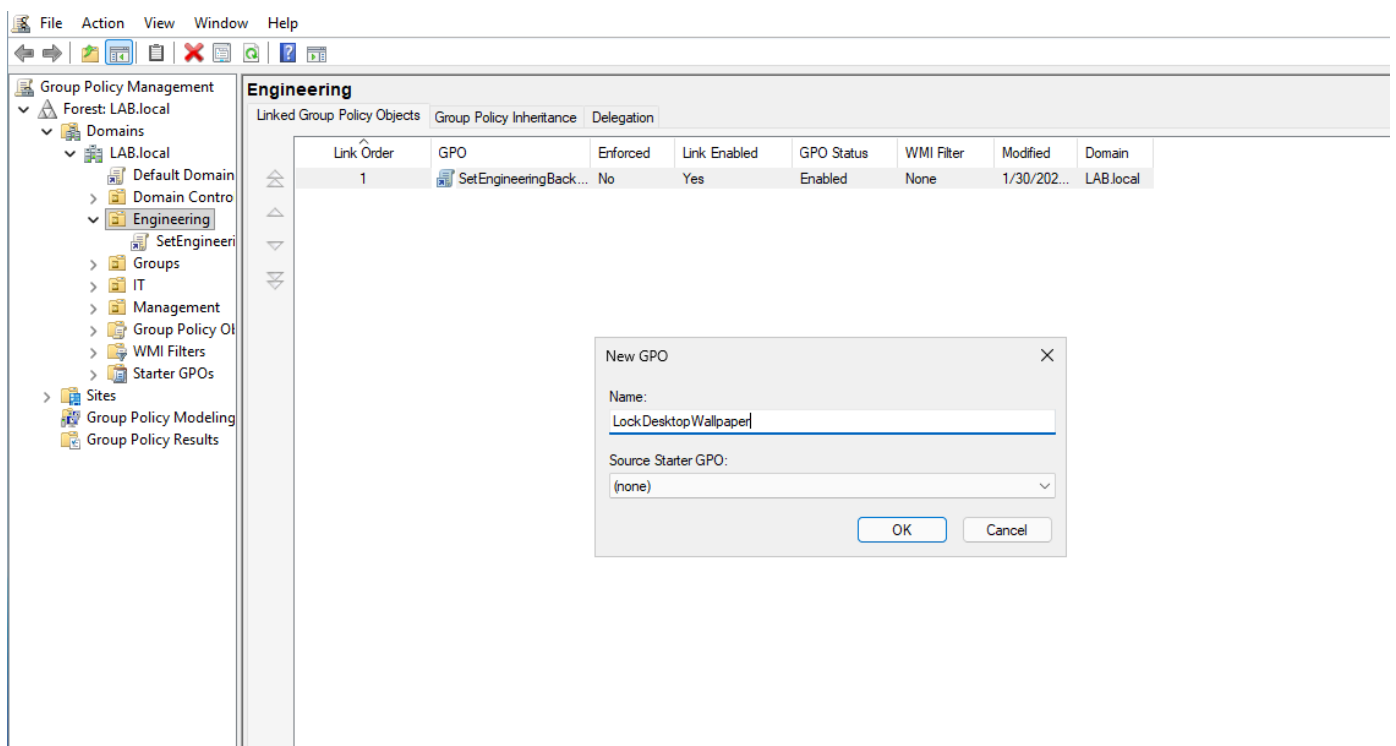
hit apply, and ok, and then close both windows.

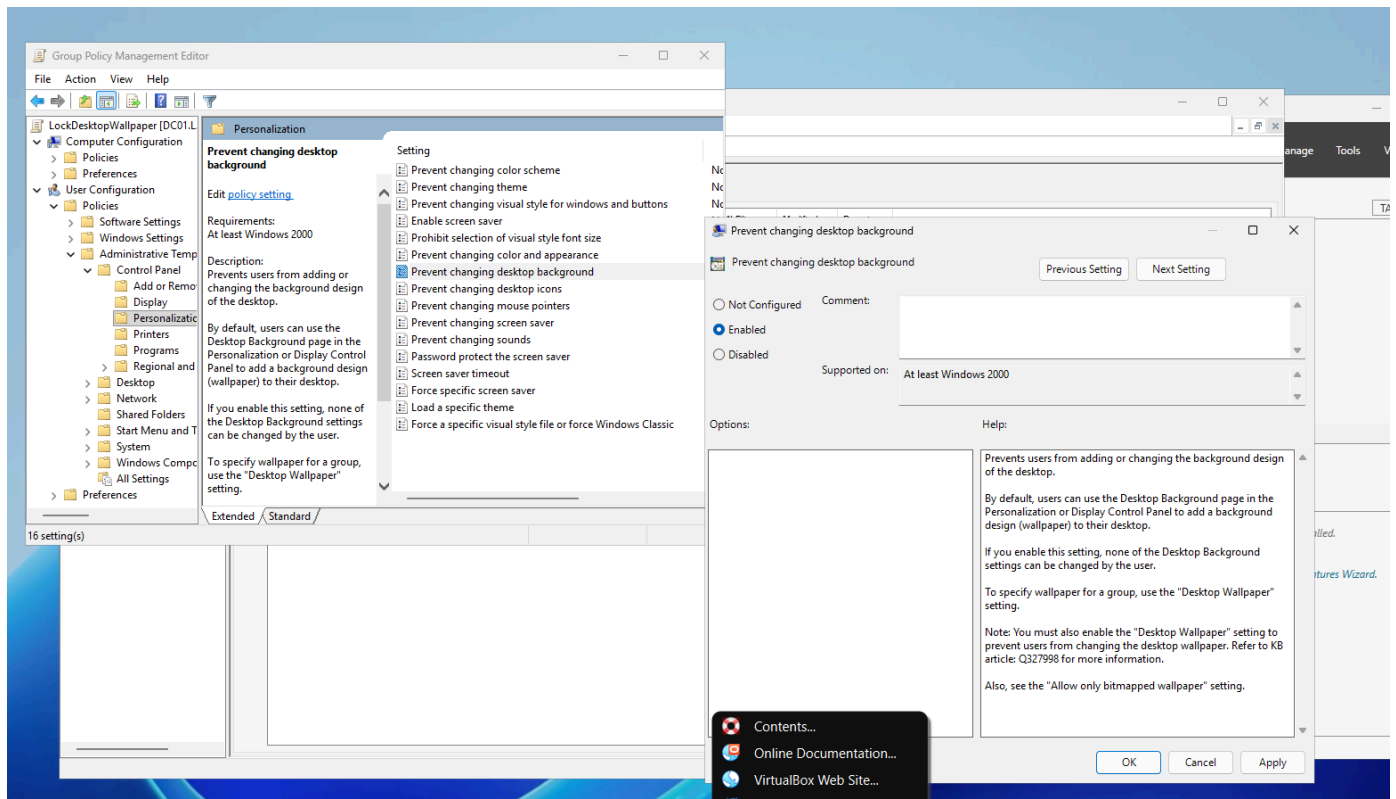
login as management member who was in the administrator group of Engineers from the win 11 ent vm and see that the wallpaper is changed.



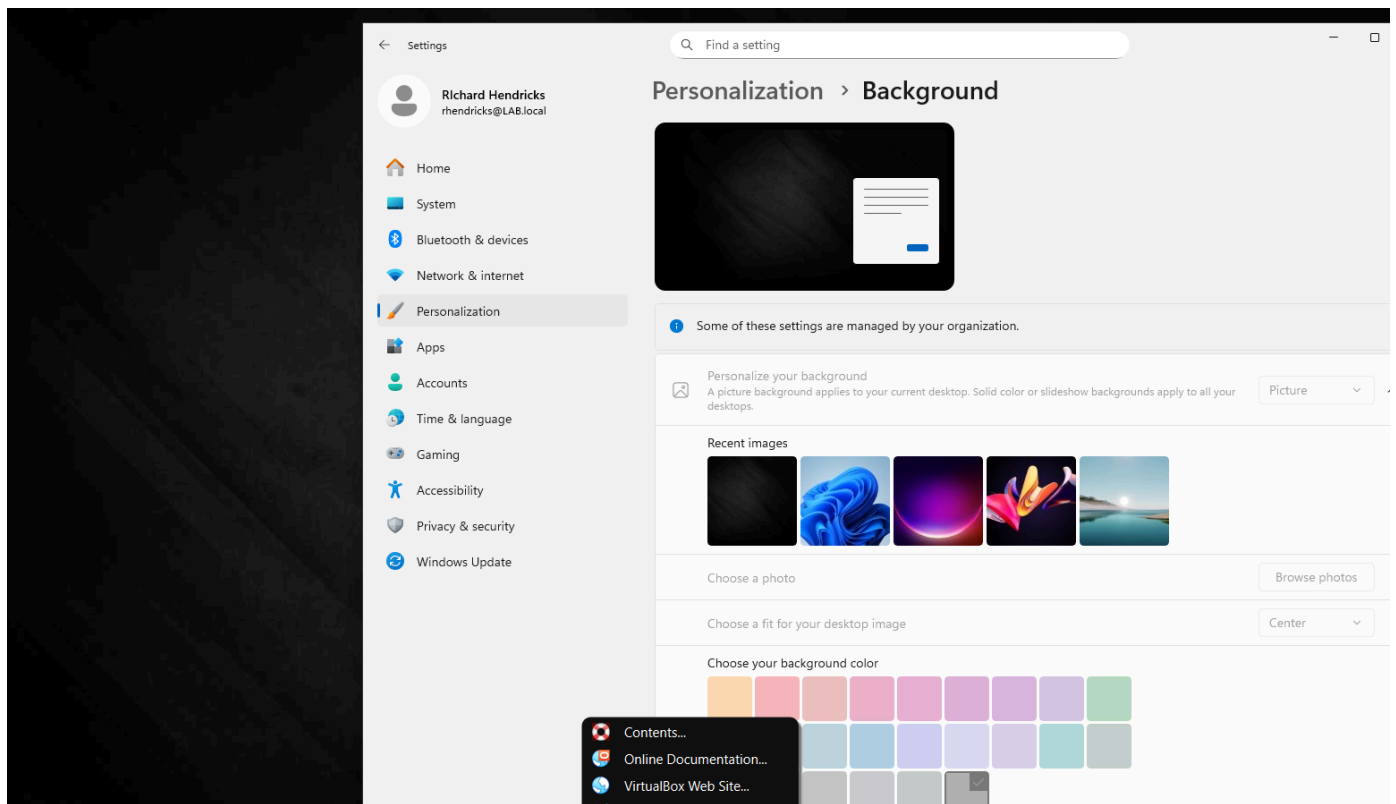
Ticket Interrupt:

not letting the engineers change their desktop backgrounds:





hit apply then ok.



Powershell:

login as administrator@lab.local from win 11 ent vm, while the server vm is logged in and running.

automating tasks with powershell,

resetting AD passwords

